Chapter-3

3.1 Classification and clinical manifestations of Dengue

The natural course of the illness:

The clinical manifestation of dengue infection varies according to the virus strain and host factors such as age, immune status, etc. Many patients infected with dengue virus remain asymptomatic. Symptoms develop after an incubation period of 4-7 days (range 3-14 days) and symptomatic patients may have one of the following conditions:

- Undifferentiated fever
- DF
- DHF
- Expanded dengue syndrome (rare)

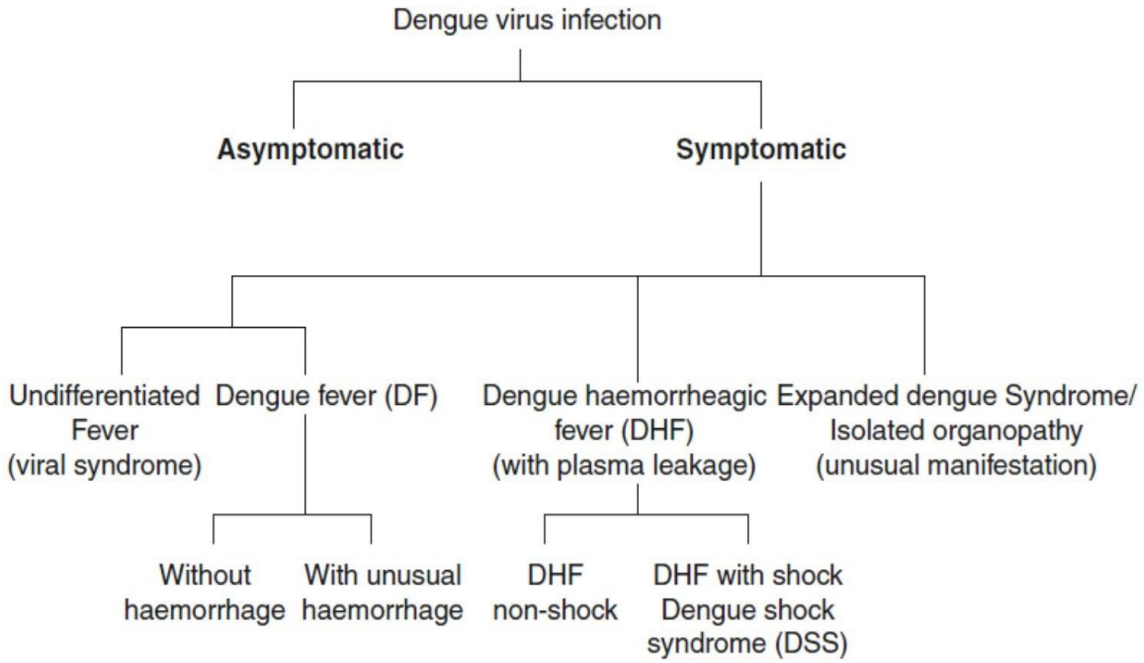


Figure 12: WHO and SEARO 2001 or WHO 1999 criteria for dengue classification. (Source: adopted from WHO criteria for dengue classification)

3.1.1 Asymptomatic Infection:

Majority of dengue virus infections are asymptomatic. However, age appears to influence the prevalence of symptomatic disease. The majority of infections in children under age 15 years are asymptomatic or minimally symptomatic.

3.1.2 Symptomatic Infection

3.1.2.1 Undifferentiated fever

Some of the patients infected with dengue virus, especially for the first time (i.e. primary dengue infection), may develop a nonspecific febrile illness with mild symptoms which are indistinguishable from other viral infections.

3.1.2.2 Classical Dengue fever (DF)

A proportion of patients infected with dengue virus develop classical dengue fever characterized by high fever, severe headache, retro-orbital pain, myalgia, arthralgia and rashes. Leucopenia and thrombocytopenia are usually observed. Although DF is usually benign, it could be an incapacitating disease due to severe headache and musculo-skeletal pains (break-bone fever). Occasionally, unusual haemorrhage such as gastrointestinal bleeding, hypermenorrhea and massive epistaxis may occur.

Fever

- Sudden onset, high grade, continued.
- The body temperature is usually between 39 °C and 40 °C (102° F to 104° F).
- The fever may be biphasic, lasting 2–7 days in the majority of cases.

Rash

first 2 to 3 days - Diffuse flushing or blanching erythema may be seen on the face, neck and chest

Third and fourth day - a conspicuous rash that may be maculo-papular or rubelliform.

Afebrile period or defervescence - Some patients have a confluent erythematous or petechial rash with small areas of normal skin, described as “isles of white in the sea of red” may appear over the dorsum of the feet, on the legs and on the hands and arms. Skin itching may be observed.

Rash in dengue fever:

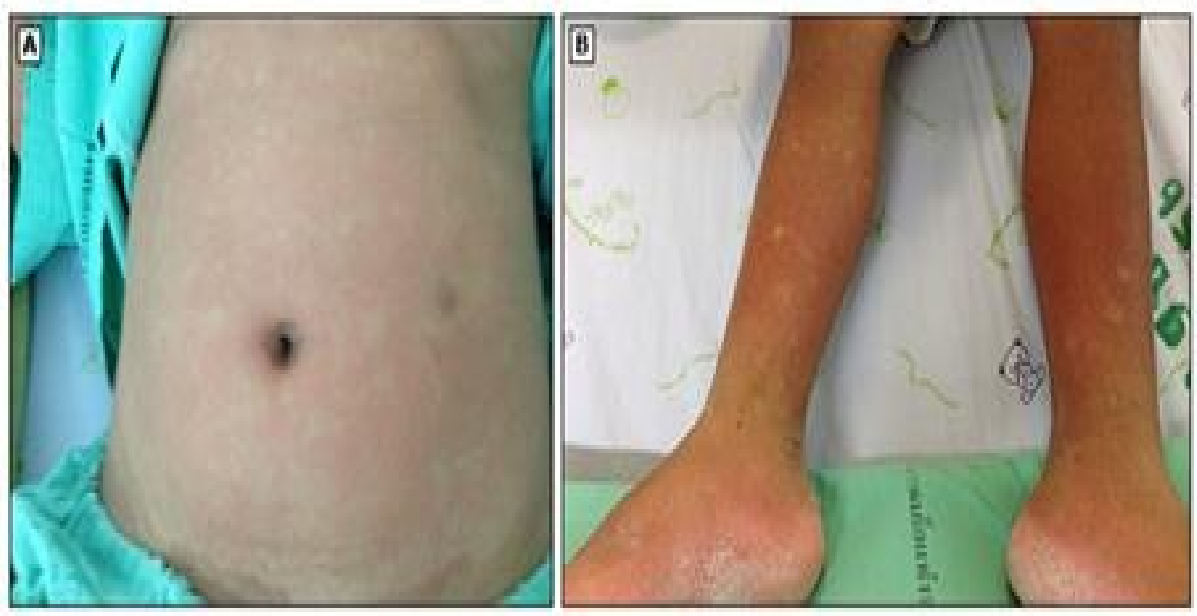


Figure 12: Dengue rash

(A) Undifferentiated macular or maculopapular rash may occur over the face, thorax, abdomen, and extremities during the acute phase of dengue. The rash is typically macular or maculopapular and may be associated with pruritus.

(B) Convalescent rash is characterized by confluent erythematous eruption with sparing areas of normal skin. It is often pruritic. The rash typically occurs within one to two days of defervescence and lasts one to five days.



Figure13: Isles of White in the Sea of Red.

Warning signs of dengue fever: (CDC 2024)

- Intense continuous abdominal pain or pain when palpating abdomen
- Persistent vomiting (≥3 episodes in 1 hr or ≥4 in 6 hrs)
- Fluid accumulation (pleural effusion, ascites, or pericardial effusion)
- Mucosal bleeding (gums, nose, vagina [metrorrhagia or hypermenorrhea], kidney [macroscopic hematuria])
- Altered mental status (irritability, drowsiness, Glasgow Coma Scale score <15)
- Hepatomegaly (≥2cm below costal margin)
- Progressive increase of the haematocrit (in at least 2 consecutive measurements taken 6 hours apart

3.1.2.3 Dengue Haemorrhage Fever (DHF)

DHF occurs in a small proportion of patients with dengue infection. Most DHF patients present with acute onset of high fever and associated symptoms similar to DF in the early febrile phase. **Plasma leakage and abnormal hemostasis** are the hallmarks of DHF which usually occurs around the defervescence (settling of fever). There is a possibility of developing shock (dengue shock syndrome) if the leakage is severe or due to significant bleeding in addition to leakage. Therefore, patients with probable dengue illness should be closely monitored as it can develop DHF with time. For efficient management of DHF it is important to understand its natural history and its dynamic nature. Clinical course of DHF is stereotypic and consists of three phases:

- Febrile phase
- Critical phase (leakage phase)
- Convalescent phase

Note: Equilibrium phase is seen in some cases after critical phase

3.2 Differential Diagnosis of Dengue Fever

Fever that has similar clinical feature of dengue like fever, rash, body ache, headache are its differential diagnosis. Clinical presentation of following infections has similarity with dengue fever: Infection by

Arboviruses: Chikungunya virus, Zika virus

Other Viral Infection: Measles, rubella and other viral exanthems, Epstein-Barr virus (EBV), Enteroviruses, Influenza, Hepatitis A, Hanta virus.

Bacterial diseases: Meningococcemia, Leptospira, Enteric fever, Melioidosis, Rickettsial disease, Scarlet fever, Sepsis.

Parasitic Disease: Malaria.

Dengue, Chikungunya and Zika

Dengue, chikungunya, and Zika are all mosquito-borne viral diseases that can cause similar symptoms, such as fever, headaches, muscle and joint pain.

Clinical signs and symptoms and laboatory findings to differentiate dengue, zika and chikungunya.

Certainty of the evidence	Signs and Symptoms		
	Dengue	Chikungunya	Zika
HIGH (findings that differentiate them)	- Thrombocytopenia - Progressive increase in hematocrit - Leukopenia	Arthralgias	Pruritus
MODERATE (findings that probably differentiate them)	- Anorexia or hyporexia - Vomiting - Abdominal pain - Chills - Hemorrhages (includes bleeding on the skin, mucous membranes, or both)	Rash Conjunctivitis Arthritis Myalgias or bone pain	Rash Conjunctivitis
LOW (findings that may differentiate them)	- Retro-ocular pain - Hepatomegaly - Headache - Diarrhea - Dysgeusia - Cough - Elevated transaminases - Positive tourniquet test	Hemorrhages (includes bleeding on the skin, mucous membranes, or both)	Adenopathies Pharyngitis or odynophagia

Guidelines for the Clinical Diagnosis and Treatment of Dengue, Chikungunya, and Zika. Washington, D.C.: Pan American Health Organization; 2022
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Table 2: Table showing clinical differentiation among dengue, chikungunya and zika.

Chapter-4

4.1 Laboratory Investigations for Dengue:

Dengue (DEN) virus, which has 4 distinct serotypes, i.e. DEN -1, DEN-2, DEN-3 & DEN-4. Early laboratory confirmation of clinical diagnosis may be important because some patients progress within a short period from mild to severe disease and sometimes to death.

4.1.1 Laboratory Tests for Diagnosis and Monitoring

The management of dengue cases is based on both clinical features and laboratory evaluations. Suggestive laboratory test of dengue cases management is following:
CBC and NS-1 Ag/ Anti-Dengue Ab should be done in all suspected dengue patients at their first visit depending on the day of presentation.

4.1.1.1 Complete Blood Count (CBC):

WBC counts: **Leucopenia** is common in both adults and children with DF (Level 1) and has an important diagnostic implication in early period. The change in total white cell count (≤5000 cells/mm^3) and ratio of neutrophils to lymphocyte (neutrophils <lymphocytes) is useful to predict the critical period of plasma leakage (Level 6). NLR (Neutrophil Lymphocyte ratio) <2 strongly correlates with low platelets indicating severity of disease and reversal of NLR >2 correlates with improvement of disease (Level 6). This finding precedes thrombocytopenia or rising haematocrit. These changes seen in DF and DHF.

Total Platelet count (PC): **Thrombocytopenia** is observed in patients with DF & DHF. The level of platelet count is correlated with severity of DHF (level 1). Thrombocytopenia (<100,000/mm^3) usually precedes/accompanies plasma leakage.

Haematocrit (Hct): Haemotacrit is most important tool in diagnosing and guiding effective treatment of dengue patient(Level 5). Approximately 5-10% increase in Hct may occur due to high fever, anorexia and vomiting. A sudden rise in haematocrit is observed simultaneously or shortly after the drop in platelet count. Rising haematocrit more than 10 % indicates starting plasma lekage and rising haematocrit by 20% from the baseline indicates established plasma leakage. For example, rising haematocrit from 35 to 42 indicates 20% rise. In critical phase / leaking phase haematocrit may not rise due to concomitant concealed or overt bleeding. During plasma leakage period Hct should be monitored at least twice daily and more frequently considering the clinical scenario, If facilities available. Bedside haematocrit is suggested in admitted patients. There is no data but it assumes that base line Hct in child, adult female and adult male are 35%, 38% and 40% respectively.

4.1.1.2 Dengue antigen and antibody tests:

Detection of antigen: dengue NS1 antigen (non-structural protein 1):

- The Dengue NS1 antigen usually positive on first day of illness.
- This antigen test may become negative from 4-5 days of illness.

> Note: The sensitivity, specificity, positive, and negative predictive value of rapid dengue NS1 antigen test were 81.5%, 66.7%, 78.2%, and 71.1%, respectively whereas that of NS1 ELISA were 89.9%, 100%, 100%, and 94%, respectively. Concordance of Rapid NS1 and NS1 ELISA with PCR was 75.5% and 94%.

Dengue IgM /IgG antibody test (MAC ELISA or enzyme-linked immuno-sorbent assay or Rapid Immnuno Chromatomatic Test):

- Anti-dengue IgM can be detected after 5 days of the onset of fever and reaches highest level after 7 days of fever.
- It can be detected up to 1-3 months after dengue fever.
- In primary dengue infection- anti-dengue IgM will be detected earlier than IgG which (IgG) will appear on the 9th or 10th day of illness. Level of this IgG may persist at low levels for life long, indicating past dengue infection.
- In secondary dengue infection- higher levels of anti-dengue IgGand lower levels of anti-dengue IgM are detected at 2nd week. The anti-dengue IgG levels detected lifelong.

4.1.1.3 Nucleic Acid Detection:

- Dengue diagnosis is confirmed by detecting DEN viral RNA using reverse transcriptase polymerase chain reaction (RT-PCR; This test is usually positive within 5 days of onset of symptoms. It can also be detected by traditional PCR.)

4.1.1.4 Dengue Virus Isolation:

- This test is usually done for research purpose and are not readily used for routine diagnostic test.
- Dengue virus isolation in cell culture (Vero, C6/36 cell lines) done from serum, plasma and leucocytes is the most definitive test for dengue infection, which can be usually accomplished in majority of cases if the sample is taken in the first five days of onset of symptoms.
- Isolation of dengue virus also from CSF or autopsy samples.
- Detection of dengue virus or antigen in tissue, serum or cerebrospinal fluid by immunohistochemistry, immunofluorescence
- Detection of dengue virus genomic sequences by reverse transcription- polymerase chain reaction.

4.1.2 Biochemical Tests:

Serum AST (SGOT) and ALT (SGPT): Transaminases are commonly abnormal in dengue fever where classically AST is more than ALT. Elevated aspartate aminotransferase levels were strongly associated with severe dengue. (Level 1)
In Classical Dengue fever AST/ ALT may be slightly raised (2-5 times of upper limit of normal) but in dengue haemorrhagic fever ALT and AST may be 5 to 15 times the upper limit of normal. Serum Albumin <3.5 gm/dl indicates plasma leakage.

Additional following test might be needed depending on the severity of clinical scenario:

- Serum Creatinine/Blood urea nitrogen (BUN) may be elevated in prolonged shock.
- Serum Electrolyte: Hyponatremia is frequently observed in DHF/DSS
- VBG (venous blood gas analysis): Metabolic acidosis is frequently found in cases with prolonged shock
- RBS may be low in DSS
- Serum calcium: Hypocalcaemia (corrected for hypoalbuminemia) has been observed in DHF.

4.1.3 Coagulation Profile:

Assays of coagulation and fibrinolytic factors show reduction in DSS cases. Activated Partial thromboplastin time (aPPT) and prothrombin time (PT) are prolonged in about half and one third of DHF cases respectively. Thrombin time (TT) is also prolonged in severe cases.

4.1.4 Other tests:

- Chest X-Ray: Pleural effusion on X-ray in dengue indicates fluid accumulation.
- Ultrasonography: To detect pleural effusions or ascites, it is essential to obtain objective evidence of selective capillary plasma leaking into the chest and abdominal cavities.
- Urine R/M/E: Albuminuria/Hematuria may be present
- Other tests might be done to exclude other diagnosis such as ICT for malaria, microscopy for malaria, Blood culture to exclude enteric fever.
- ECG, Echocardiography, Troponin I, CK-MB if myocarditis is suspected.
- CSF: In dengue encephalitis.

N.B: It should be noted that the use of medications such as analgesics, antipyretics, anti- emetics and antibiotics can interfere with liver function and blood clotting.

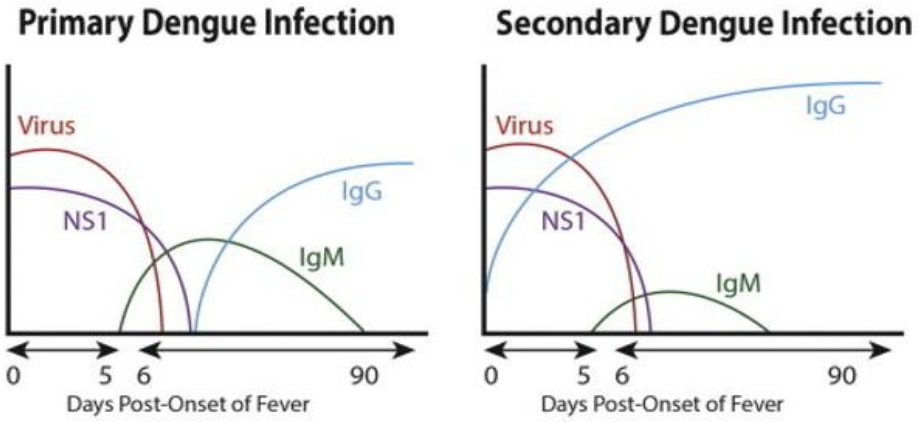


Figure 19: Changes in serological markers used in dengue diagnosis during primary and secondary dengue infections

4.1.5 Time and frequency of investigation:

Within 5 days: CBC, Haematocrit, Dengue NS1 antigen, SGOT, SGPT

- These tests should be done during first consultation to get the baseline characteristics like Haematocrit and complete blood count if the patient presented within 5 days of fever.
- Follow up testing may be done on 1st afebrile day, but should be done daily once DHF is suspected.
- A regular haematocrit is more important for management than the platelet count
- Even in severe dengue especially with shock haematocrit to be monitored 3 hourly or more frequently is crucial for management. (According to monitoring chart; Annexure)
- Once the platelet count begins to rise and reaches ≥ 50,000/mm^3,daily lab evaluations may be discontinued.

4.1.6 Role of point of care (POC) ultrasound in dengue:

Point of care ultrasonography is not essential to manage dengue patient but it has now evolved into an important and reliable method of diagnosing plasma leakage objectively in clinically suspected DHF patients. (Level 4)

There are many advantages of POC ultrasound. It can be done at the bedside on unprepared patients and can be repeated frequently if needed. POC ultrasound can also be used to predict the probable duration of the critical phase.

Areas of interest of POCS in dengue

Main areas:

- Gallbladder - gallbladder wall thickening and appearance of para cholecystic fluid
- Fluid in the hepato-renal pouch
- Fluid in right pleural cavity
- Fluid in the pelvis (recto vesical pouch)

Chapter-5

5.1 Dengue Case Definition

Suspected case: Acute febrile illness meeting clinical dengue criteria without confirmatory tests.

Probable Case: Acute febrile illness with serological diagnosis.

Confirmed Case: Laboratory confirmation through any one of the followings

- Positive RT-PCR or NS1 antigen test
- Positive immunohistochemistry from necropsy tissues
- Seroconversion (negative to positive IgM/IgG in paired sera)
- Fourfold rise in IgG in paired sera samples.

Severity Classification:

Dengue without warning signs: Fever (2-7 days) with two or more of the followings; nausea, vomiting, rash, aches and pains, positive tourniquet tests, or leucopenia (WBC <5000/mm^3)

Dengue with warning signs: Dengue fever with any one of the warning signs. **(page 28)**

Severe Dengue:

- Severe plasma leakage leading to shock (DSS) or fluid accumulation causing respiratory distress.
- Severe bleeding as per clinical evaluation
- Severe organ involvement (e.g. elevated AST/AST≥ 1000iu/L, impaired consciousness, or organ failure.

5.2 Outpatient Assessment and Management of a Patient with Dengue Fever.

Out-Patient Department:

Fever Corner - A triage center for febrile patients

Fever corner should consist of a **Triage Desk** where fever patients will be registered and assessed based on duration of fever.

- Then the patients can be seen by medical officers and managed accordingly.

In the present hyper-endemic setting in Bangladesh, dengue illness (DF and DHF) should be considered in any patient presenting with an acute febrile illness. Clinical evaluation of dengue is based on symptoms and signs. Basic investigations depend on day of illness. CBC and NS1 Ag up to day 5 and CBC and Dengue Antibody from day 6 onward will support the diagnosis.

Stepwise approach on outpatient management on dengue History:

- Date of onset of fever/ illness
- Date and time of last episode of fever
- Oral intake
- Warning signs
- Diarrhoea
- Bleeding including menstrual bleeding
- Change in mental status /seizure /dizziness
- Urine output (frequency, volume and time of last voiding)
- Other important relevant histories: drug history, comorbidities.

Physical examination:

- Mental status
- Hydration status
- Haemodynamic status
- Tachypnoea/ acidotic breathing/ pleural effusion
- Abdominal tenderness/ hepatomegaly/ ascites
- Bleeding manifestation

Investigation:

- CBC and Hct
- NS-1
- SGPT/SGOT
- CRP (level 4)

Besides checking clinical and lab parameter of dengue, proper history to be taken on comorbid condition, high risk for severe disease and adverse social condition of the patients.

Patients at high risk for more severe disease and its complications may need admission even without the above criteria.

- Pregnant women
- Elderly
- Obesity
- Patients on anticoagulants

Patients with comorbid conditions:

- Diabetes mellitus
- Hypertension
- Chronic renal failure
- Established liver disease
- Ischemic heart disease
- Solid organ/bone marrow transplant
- Active malignancies
- Hemolytic diseases and other haemoglobinopathies
- Peptic ulcer disease
- Patients on steroid or NSAID treatment
- Patients on anticoagulants

Patients with adverse social circumstances

- Age <1 year or >65 years
- Living alone or has poor access to healthcare facilities
- Lack of transportation
- Unstable housing
- Extreme poverty

All patients should be assessed in a stepwise manner to reach at the diagnosis, determine the severity and place of management.

At the end of evaluation, attending physician should be able to assess the patient on following points.

- Could this patient have dengue?
- Are there any warning sign?
- Is the patient having plasma leakage?
- Is the patient in shock?
- What is the haemodynamic and hydration status of the patient?
- Are there any co-morbidities or high-risk conditions or adverse social condition?
- Is hospitalization required immediately/early?

Fever	Warning Signs (WS)	Comorbid condition/High risk for severe disease/Adverse Social condition	Vital Signs (VS) & Urinary output (UOP)	Fresh CBC Finding	Patient Risk Status at the Time of Assessment	Hospitalization
Short duration of fever (or fever within last ≤48 hours)	Absent WS	Absent	Normal VS & Normal UOP	WBC low (Leucopenia) Platelet ≥100x10^9/L Hct: 35-40% or baseline	No immediate risk	Home management as well as followed up and assessed regularly.
	Present WS	Present	Derange VS Reduced UOP	WBC: decreasing Platelet ≤100x10^9/L Increasing Hct: Leaking Decreasing Hct: Bleeding	High risk.	Inpatient management

Table 7: Showing dengue patients who need inpatient/outpatient management.

Note:
➢ It is important to understand that vital signs and other parameters may be influenced by comorbidities and high-risk conditions (e.g., urine output in diabetes).
➢ Settling of fever is not a sign of recovery. Some patients will start plasma leakage around defervescence.

Home management of patients:

Patients without warning signs or coexisting conditions or social risk should get home based care through a register of dengue patient.

Recommended measures in home management:

- Adequate physical rest in comfortable clothing, and preferably under a mosquito net
- Control of temperature
  - Tepid sponging for fever
  - **Do not exceed 3 gm/day of paracetamol. (60mg/kg/24hrs in child)**
  - Warn the patient that the fever may not fully settle with paracetamol and advise not to take excess.
- Adequate Hydration
  - Ensure adequate oral fluid intake of around 2400 ml for 24 hours (if the body weight is less than 50kg, give fluids as 2ml/kg/hour for 24 hours) and it is better to spread this amount of fluid across the day.
  - This should consist of fluid such as ORS, green coconut water, other fruit juices or soup which has electrolytes rather than plain water
- Ensure adequate urine output. (Encourage to pass urine every 3-4 hours). It is useful to measure the urine output, if possible, and document with time, as any reduction in urine output can be easily identified.
- Patient should be informed about warning sign. **(page 28)**

- If the appetite is good, patients encouraged to take light and nutritious food.
- If the patient's appetite is poor, forced feeding should not be done, as taking adequate amounts of liquid is sufficient during illness.
- It is advisable to avoid red/ brown (i.e. dark) colored food (red dragon fruit, beet root)/ beverages to avoid confusion with blood-stained vomitus/ melaena)
- Anti-emetics and PPI/H2 receptor blockers if necessary.
- Avoid all NSAIDS (including COX-2 inhibitors) and steroids.
- Temporally withhold warfarin, aspirin, clopidogrel & dipyridamole, in patients who take these on long term basis. Take expert opinion where these drugs are necessary.

Follow up plan of home management

- Review daily with CBC. First CBC should be done at least at the beginning of the third day of fever/illness (preferably with SGOT and SGPT).
- These patients should be followed up until the patient become afebrile for 24-48 hours with stable clinical parameters. Usually, by seven days, most patients recover.
- If any indication for hospitalization appears within this follow up period, patient should get admitted to hospital.

Note: Written instruction to the patient/carer giver will be provided in Bengali regarding management of the patient. Instructions will be given in Bengali about investigations timing and how and when they can communicate with physician either physically or phone call/WhatsApp. (Instructions of patient/caregiver is in annexure).

Indications for immediate return for review / hospitalization

Advise patients to return to the hospital immediately, if they develop any of the following conditions:

- Clinical deterioration with settling of fever
- Inability to tolerate oral fluids
- Severe abdominal pain
- Lethargy or irritability / restlessness

Bleeding tendency like abnormal vaginal bleeding, blood vomiting, black tarry stool, bleeding from nose and gum.

Reduced urine output for the last 4-6 hours (<150-200 ml/ 4-6 hours if weight is 50kg or more)

- Cold and clammy extremities
- Difficulty in breathing.

Before transferring a patient from OPD to ward or another hospital after initial evaluation:

Following measures to be taken strictly

- Patient should be stabilized.
- All clinical findings, investigation results and treatment given should be clearly documented and sent with the patient.
- Reasons for transfer should be clearly documented in the transfer form
- Before transferring, the patient's condition should be briefed with the relevant management team of the hospital where the patient is being transferred.

Chapter -4

4. Diagnosis and assessment of patients with hypertension

4.1 Diagnosis

The diagnosis of hypertension should be based on multiple blood pressure measurements, taken on separate occasions over a period. BP can be highly variable; thus, the diagnosis of hypertension should not be based on a single set of BP readings at a single office visit. In general, the diagnosis of hypertension should be based on at least three blood pressure measurements per visit and at least two to three visits at 1–4-week intervals (depending on the BP level) to confirm the diagnosis of hypertension[9].

The diagnosis might be made on a single visit, if BP is ≥160/100 mmHg or there is accompanying evidence of hypertension mediated organ damage (HMOD) (e.g., hypertensive retinopathy with exudates and hemorrhages, or LVH, or vascular or renal damage)[10]. When the initial SBP is between 140 and 159 mmHg, or the DBP is between 90 and 99 mmHg, repeat measurements should be performed on three separate occasions within a period of 1 to 4 weeks to determine whether a diagnosis of hypertension is valid. All measurements should be taken in the same arm. If possible and available, the diagnosis of hypertension should be confirmed by out-of-office BP measurement (i.e., HBPM and/or ABPM)[9].

In the initial evaluation, measurement of blood pressure should be performed in both arms. If the difference in readings between arms is more than 10 mmHg, repeat the measurements at 1-2 mins interval. If the difference in readings between arms remains more than 10 mmHg on the second measurement, measure subsequent blood pressures in the arm with the higher reading and it should be noted. If the difference is >20 mmHg, consider further investigation[9].

If hypertension is not diagnosed, measure the person's clinic blood pressure at least every 3 years subsequently, and consider measuring it more frequently if the person's clinic blood pressure is close to 140/90 mmHg (Table 3).

Table 3. Blood Pressure Measurement Plan According to Office Blood Pressure Levels[10]		
Office Blood Pressure Levels (mmHg)		
<130/85	130–159/85–99	>160/100
Remeasure within 3 years (1 year in those with other risk factors)	Confirm with repeated office visits. If possible, confirm with out-of-office BP measurement	Confirm within a few days or weeks

Postural Hypotension[16]:

Patient should lie down for 5 minutes, and measure BP and pulse rate. Then have the patient stand and repeat BP and pulse rate measurements 1 and 3 minutes after standing. A drop in SBP of ≥20 mmHg, or in DBP of ≥10 mmHg, or experiencing lightheadedness or dizziness on standing is considered abnormal and indicates postural hypotension.

Measure standing blood pressure in treated hypertensives with symptoms of postural hypotension (falls or postural dizziness) and at the first visit in the elderly and people with diabetes. BP is to be measured first in supine or seated position, and again with the person standing for at least 1 minute prior to measurement. If the systolic BP falls by ≥ 20 mmHg when the person is standing, medication is to be reviewed, subsequent BP is to be measured with the person standing and the patient may be referred to specialist care if symptoms of postural hypotension persist.

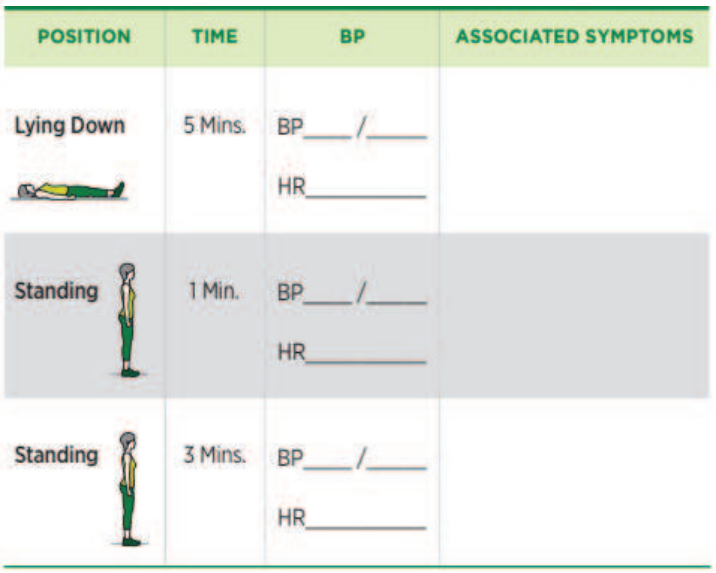


Figure 3: Assessing Postural Hypotension

4.2 Clinical evaluation and assessment of hypertension

4.2.1 Clinical evaluation:

The goals of the initial evaluation of the hypertensive patient are to

1. Establish the diagnosis and grade of hypertension.
2. Screening for potential secondary causes of hypertension.
3. Identify factors that potentially contribute to development of hypertension (lifestyle, concomitant medications, or family history).
4. Identify concomitant CV risk factors (including lifestyle and family history) (Table 4)
5. Identify concomitant diseases and establish whether there is evidence of HMOD or existing cardiovascular, cerebrovascular, or renal disease. (Table 5)

These goals are usually accomplished by a thorough medical history, physical examination, and simple laboratory investigations. While waiting for confirmation of a diagnosis of hypertension, carry out some routine investigations for assessment of HMOD. If hypertension is not diagnosed but there is evidence of HMOD, consider carrying out investigations for alternative causes of HMOD.

History

Most patients with hypertension are asymptomatic, the high blood pressure usually having been noted during an incidental clinical examination. A proportion of patients will present with a major complication of hypertension such as stroke or myocardial infarction, but only a small number will present with symptoms directly attributable to hypertension such as breathlessness or headache. That is why hypertension is called a 'Silent killer'. The key issues that need to be addressed in the history include:

Risk factors assessment:

* Family and personal history of hypertension, CVD, stroke, or renal disease.
* Family and personal history of associated risk factors (e.g., familial hypercholesterolaemia)
* Dietary history and salt intake
* Smoking history
* Alcohol consumption
* History of physical exercise/sedentary lifestyle
* History of erectile dysfunction

- Sleep history, snoring, sleep apnoea (information also from partner)
- Previous hypertension in pregnancy/pre-eclampsia
- History of oral contraceptive use.

History of symptoms of HMOD, CVD, stroke, and renal disease

- For Brain and eyes: headache, vertigo, syncope, impaired vision, TIA, sensory or motor deficit, stroke, carotid revascularization, cognitive impairment, dementia (in the elderly).
- Heart: chest pain, shortness of breath, oedema, myocardial infarction, coronary revascularization, syncope, history of palpitations, arrhythmias (especially AF), heart failure
- Kidney: thirst, polyuria, nocturia, haematuria, urinary tract infections.
- Peripheral arteries: cold extremities, intermittent claudication, pain free walking distance, pain at rest, peripheral revascularization, patient or family history of CKD (e.g., polycystic kidney disease).

History of Antihypertensive Drug Treatment

- Current/past antihypertensive medication including effectiveness, intolerance or side effects of previous medications.
- Adherence to therapy and in non-compliant patients the reason behind it (including financial constraint).

Physical Examination

The physical examination should include the following:

- Weight and height measured on a calibrated scale, with calculation of BMI
- Waist circumference
- Neurological examination and cognitive status
- Fundoscopic examination for hypertensive retinopathy
- Palpation and auscultation of heart and carotid arteries
- Palpation of peripheral arteries
- Comparison of BP in both arms (at least once)
- Skin inspection: cafe-au-lait patches of neurofibromatosis (phaeochromocytoma)
- Kidney palpation for signs of renal enlargement in polycystic kidney disease

- Auscultation of heart and renal arteries for murmurs or bruits indicative of aortic coarctation, or renal artery stenosis
- Comparison of radial with femoral pulse: to detect radio-femoral delay in aortic coarctation
- Signs of Cushing's disease or acromegaly
- Signs of thyroid disease

Table 4: Major risk factors for cardiovascular diseases

- Levels of SBP and DBP
- Tobacco use
- Dyslipidemia
  - Total cholesterol >5.1 mmol/L
  - LDL >3 mmol/l, OR
  - HDL <1 (men) and <1.2 mmol/l (women)
- Diabetes mellitus
- Age (>55 years for men, >65 years for women)
- Family history of premature cardiovascular diseases
  - (men <55 years or women <65 years)
- Central obesity
  - (Waist circumference >90 cm for men, >80 cm for women)

Table 5: Manifestations of hypertension mediated organ damage (HMOD)

Organ system	**Manifestations & their suggestive signs**
Cardiac	Left ventricular hypertrophy, coronary heart disease, heart failure ➢ forcible and displaced apex beat ➢ accentuation of the aortic component of the second heart sound ➢ abnormal cardiac rhythms ➢ S4 (decreased ventricular compliance) ➢ ventricular gallop ➢ pulmonary rales or crackles
Cerebrovascular	Transient ischaemic attack, stroke ➢ carotid bruits; motor or sensory defects;
Peripheral vasculature	➢ Absence of one or more major pulses in extremities (except dorsalis pedis) with or without intermittent claudication
Renal	➢ GFR <60 ml/min/1.73 m^2 ➢ proteinuria (1+ or greater) ➢ microalbuminuria (2 out of 3 positive tests over a period of 4-6 months) ➢ Dependent (leg) edema
Retinopathy	Fundoscopic abnormalities - Haemorrhages or exudates, with or without papilloedema

4.2.2 Investigations

All hypertensive patients should undergo a limited number of investigations. However, when starting pharmacological therapy for hypertension, WHO suggests obtaining tests to screen for comorbidities and secondary hypertension, but only when testing does not delay or impede starting treatment. In low-resourced areas or non-clinical settings, where testing may not be possible because of additional costs, and lack of access to laboratories and ECG, treatment should not be delayed, and testing can be done subsequently. Some medicines, such as long-acting dihydropyridine calcium-channel blockers (CCBs) are more suitable for initiation without testing, compared to diuretics or angiotensin-converting enzyme inhibitors (ACEi)/angiotensin-II receptor blockers (ARBs)[12].

All hypertensive patients should undergo a limited number of investigations, these include: -

- Blood test
    - Sodium, potassium, serum creatinine and estimated glomerular filtration rate, (eGFR)
    - If available lipid profile and fasting glucose
- Urine analysis: Dipstick urine test
- 12-lead ECG for detection of atrial fibrillation, left ventricular hypertrophy (LVH), ischemic heart disease.

For most patients: -

- Haemoglobin and/or haematocrit
- Chest X-ray: to detect cardiomegaly, heart failure, coarctation of the aorta
- Echocardiogram: to detect or quantify left ventricular hypertrophy

The nature and scale of further investigations will be determined by the index of suspicion of a secondary cause for hypertension and assessment for HMOD

Tests for assessment for HMOD

- Urine albumin: creatinine ratio: To detect elevations in albumin excretion indicative of possible renal disease.
- Blood creatinine and eGFR: To detect possible renal disease.
- Fundoscopy: To detect hypertensive retinopathy, especially in patients with grade 2 or 3 hypertension.

- Echocardiography: To evaluate cardiac structure and function when this information will influence treatment decisions.
- Carotid ultrasound: To determine the presence of carotid plaque or stenosis, particularly in patients with cerebrovascular disease or vascular disease elsewhere.
- Abdominal ultrasound and Doppler studies:
    i) To evaluate renal size and structure (e.g., scarring) and exclude renal tract obstruction as possible underlying causes of CKD and hypertension
    ii) Evaluate abdominal aorta for evidence of aneurysmal dilatation and vascular disease.
    iii) Examine adrenal glands for evidence of adenoma or phaeochromocytoma (CT or MRI preferred).
    iv) Renal artery Doppler studies to screen for the presence of renovascular disease, especially in the presence of asymmetric renal size
- Pulse Wave velocity (PWV): An index of aortic stiffness and underlying arteriosclerosis
- Ankle Branchial Index (ABI) Screen for evidence of Lower Extremity Artery Disease (LEAD)
- Cognitive function testing: To evaluate cognition in patients with symptoms suggestive of cognitive impairment.
- Brain imaging: To evaluate the presence of ischaemic or haemorrhagic brain injury, especially in patients with a history of cerebrovascular disease or cognitive decline.

5.3.3 National protocol for management of hypertension at primary care level

An expert group convened by Non communicable Disease Control Programme of Directorate General of Health Services has developed a protocol for diagnosis and management of hypertension at primary care level in 2017 (Fig 7)[14.] Currently, an adapted version of this protocol is being used at NCD corners of all Upazila Health complexes (Fig. 8).

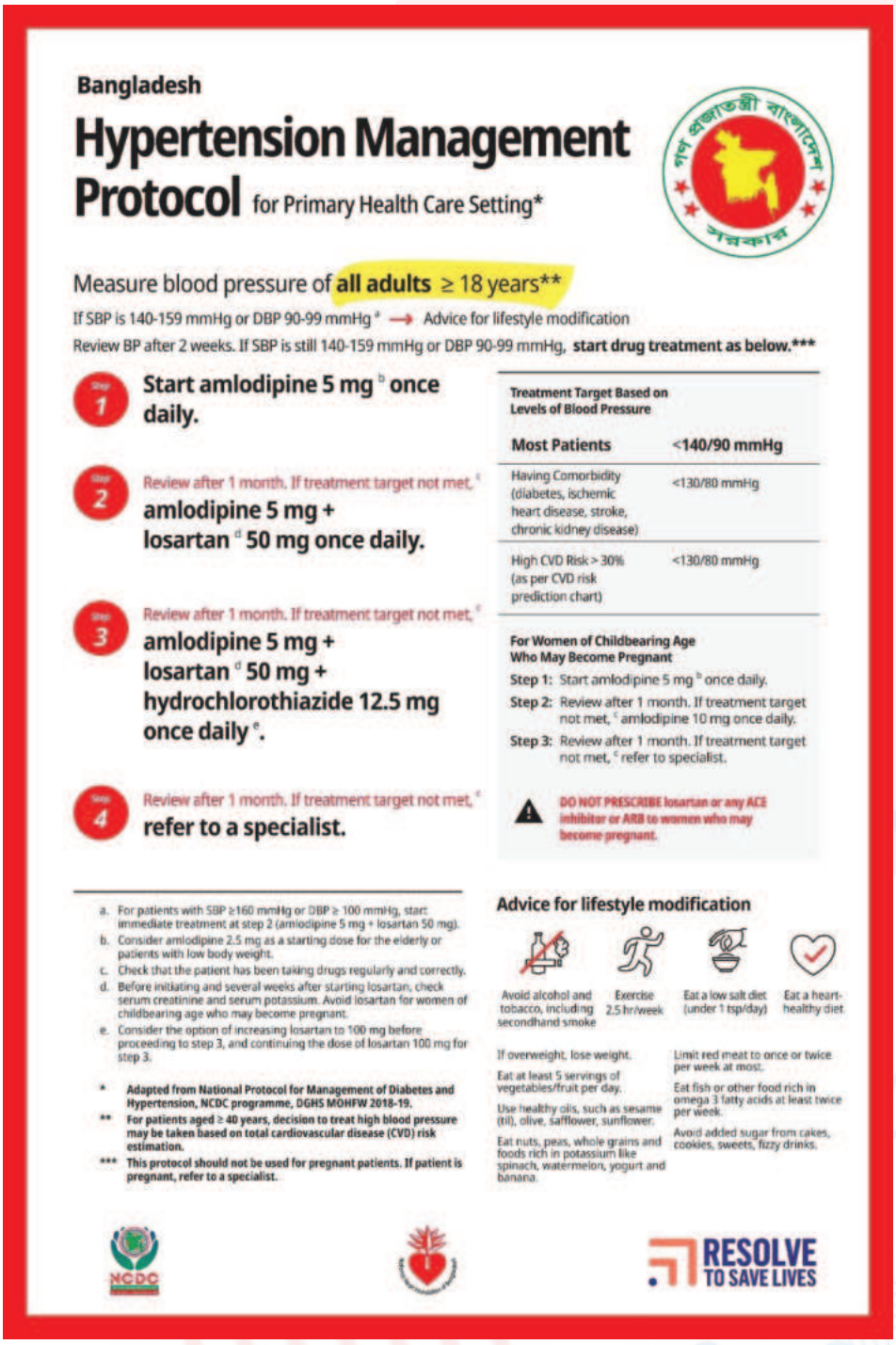


Figure 8: Hypertension Management Protocol at Primary Health Care Settings adapted from NCDC, DGHS, MOHFW

5.3.4 Frequency of assessment and follow-up visits

A monthly follow up after initiation or a change in antihypertensive medications until patients reach target should be done. Once target BP is achieved, follow-up at three to six-month interval is appropriate. Recommended blood pressure measurement plan according to office blood pressure levels is as shown in Table 6.

Table 6: Recommendations for follow-up based on initial blood pressure measurements for adults

Initial BP (mmHg) Systolic and Diastolic	Follow-up recommended to confirm diagnosis and/or review response to treatment
<130 and <85	Remeasure within 3 years (1 year in those with other risk factors)
130-159 and/or 85-99	Confirm with repeated office BP measures within 1- 4 weeks
≥160 and/or 100	Evaluate and Initiate treatment immediately

5.3.5 Goals for BP lowering treatment

Efforts must be made to reach the target BP level. These targets should be reached within 3 months. In general, once the BP is controlled, most patients will require life-long treatment.

- A target blood pressure treatment goal of <140/90 mmHg in all patients with hypertension without comorbidities.
- A target systolic blood pressure treatment goal of <130 mmHg in high-risk patients with hypertension (those with high CVD risk, known CVD, diabetes mellitus, chronic kidney disease)[10].

5.3.6 When to Refer

Patients with the following conditions should be referred to the appropriate specialist for further assessment. Indications for referral to the appropriate specialist include:

- accelerated or malignant hypertension
- suspected secondary hypertension
- resistant hypertension
- recent onset of target organ damage
- pregnancy
- children <18 years old

Cholera patient treatment flowchart

Case definition suspected cholera: *Outside of cholera outbreak:* any patient ≥2 years presenting with acute watery diarrhoea and severe dehydration or dying from acute watery diarrhoea. *Where a cholera outbreak is declared:* any person presenting with or dying from acute watery diarrhoea.

START: Does the patient fit the case definition?

No → Manage outside of cholera pathway

Yes → **Assess for shock / severe dehydration**

Patient has one or more danger sign(s):
- Lethargic or unconscious
- Absent or weak pulse
- Respiratory distress

OR patient has at least 2 of the following:
- Not able to drink or drinks poorly
- Skin pinch goes back very slowly
- Sunken eyes

Severe dehydration or shock → **PLAN C**

Give intravenous **Ringer's lactate**…

Age	Ringer's lactate (mL)
under 1 year	30 mL / kg in 60 minutes, then 70 mL / kg over the next 5 hours
1 year +	30 mL / kg in 30 minutes, then 70 mL / kg over the next 2½ hours

Be alert for hypoglycaemia, and treat rapidly with intravenous dextrose

Reassess every 15–30 minutes
- Awake and alert ***AND***
- Able to drink ***AND***
- Improved pulse strength

Yes → *Some dehydration* → **PLAN B**

No → **Assess for some dehydration**

Patient has at least 2 of the following:
- Irritable or restless
- Sunken eyes
- Rapid pulse
- Thirsty (drinks eagerly)
- Skin pinch goes back slowly

Give oral rehydration solution (**ORS**) **over 4 hours, 75 mL/kg:**

Age	Weight (kg)	ORS (mL)
< 4 months	*under 5 kg*	200–400
4–11 months	*5–7.9 kg*	400–600
12–23 months	*8–10.9 kg*	600–800
2–4 years	*11–15.9 kg*	800–1200
5–14 years	*16–29.9 kg*	1200–2200
15 years +	*30 kg +*	2200–4000

Reassess at least every hour

Severe dehydration ***OR*** Severe vomiting → Yes

Awake and alert ***AND*** Able to drink ***AND*** Improved pulse strength → Yes → **PLAN A**

No → *No dehydration*

No hydration found
- Awake and alert
- Normal pulse
- Normal thirst
- Eyes not sunken
- Skin pinch normal

PLAN A

Give oral rehydration solution (**ORS**) **over 4 hours** according to age:

Age	ORS (mL)
< 24 months	50–100
2–9 years	100–200
10 years +	As much as wanted

Reassess and consider discharge if
- Has no signs of dehydration ***AND***
- Takes ORS without vomiting ***AND***
- Has no watery stools for 4 hours ***AND***
- Is able to walk without assistance ***AND***
- Is passing urine

Reference (scan QR code)

1) Global Task Force for Cholera Control (GTFCC)

Antibiotics for all patients with severe dehydration, pregnant women and the elderly:
- *Preferred is doxycycline single dose (200 mg for under 12 years, or 300 mg for older children and adults.*
- *Alternative is azithromycin single dose (20 mg / kg for under 12 years, or 1 g for older children and adults.*

World Health Organization

Evaluating dehydration and admission criteria

	Signs	Treatment	Admit
Severe dehydration	**One or more danger signs:** • Lethargic or unconscious • Absent or weak pulse • Respiratory distress **OR at least 2 of the following:** • Sunken eyes • Not able to drink or drinks poorly • Skin pinch goes back very slowly	Plan C	Yes
Some dehydration	**No danger signs AND at least 2 of the following:** • Irritable or restless • Sunken eyes • Rapid pulse • Thirsty (drinks eagerly) • Skin pinch goes back slowly	Plan B	Yes
No dehydration	• Awake and alert • Normal pulse • Normal thirst • Eyes not sunken • Skin pinch normal	Plan A	No

Treatment for **Severe dehydration**

Start intravenous fluids immediately

Plan C

Age	first give **30 mL / kg in**	then give **70 mL / kg in**
Infants *(< 12 months)*	1 hour	5 hours
All others *(12 months +)*	30 minutes	2½ hours

Start intravenous fluid immediately
Ringers lactate is the first choice

Use large cannulas
or put 2 IV lines if large volumes are to be administered rapidly

Write the start time on the IV bag
to help track how quickly fluids are being given

Reassess
at least every 15 to 30 minutes

Watch for signs of hypoglycaemia
and treat rapidly

Give ORS to replace ongoing losses
as soon as the patient can drink safely

Zinc - give for 10 days
In patients under 6 months, 10mg once per day
In patients over 6 months, 20mg once per day

World Health Organization

Treatment to
Maintain hydration

Plan C (ORS)

Age	ORS after each loose stool
< 24 months	50–100 mL
2–9 years	100–200 mL
10 years +	As much as wanted

World Health Organization

Treatment for
Some dehydration

Plan B (ORS)

1 Give ORS
over 4 hours

2 Reassess
at least every hour

3 Give additional ORS
to replace ongoing losses in addition to fluids in the table

Age	Weight (kg)	ORS in 4 hours
< 4 months	< 5 kg	200–400 mL
4 – 11 months	5 – 8 kg	400–600 mL
12 – 23 months	8 – 12 kg	600–800 mL
2 – 4 years	12 – 16 kg	800–1200 mL
4 – 14 years	16 – 30 kg	1200–2200 mL
15 + years	30+ kg	2200–4000 mL

Zinc - give for 10 days
In patients under 6 months, 10mg once per day
In patients over 6 months, 20mg once per day

Antibiotics
All pregnant women and elderly patients should receive antibiotics regardless of their hydration status or treatment plan

World Health Organization

Treatment to
Maintain hydration

Plan B

Age	ORS after each loose stool
< 24 months	50–100 mL
2–9 years	100–200 mL
10 years +	As much as wanted

World Health Organization

Treatment for
No sign of dehydration:

Plan A (ORS)

Age	ORS after each loose stool	Give enough ORS packets for
< 24 months	50 – 100 mL	500 mL / day
2 – 9 years	100 – 200 mL	1 L / day
10 years +	As much as wanted	2 L / day

Zinc - give for 10 days

In patients under 6 months, 10mg once per day
In patients over 6 months, 20mg once per day

Antibiotics

All pregnant women and elderly patients should receive antibiotics regardless of their hydration status or treatment plan

World Health Organization

Cholera fluids quick reference

Children >1 year old, and adults

(plan C is 100mL/kg in **3 hours**)

Weight (kg)	Total volume	30 mL/kg* bolus Ringers over 30 mins		70 mL/kg Ringers over 2½ hours
10–13 kg	1000 mL	250 mL in 30 mins	(≈ 8 mL/min)	750 mL in 2½ hours
13–15 kg	1200 mL	300 mL in 30 mins	(≈ 10 mL/min)	900 mL in 2½ hours
15–17 kg	1500 mL	300 mL in 30 mins	(≈ 10 mL/min)	1200 mL in 2½ hours
17–19 kg	1700 mL	400 mL in 30 mins	(≈ 15 mL/min)	1300 mL in 2½ hours
19–20 kg	2000 mL	600 mL in 30 mins	(≈ 20 mL/min)	1400 mL in 2½ hours
21–23 kg	2200 mL	700 mL in 30 mins	(≈ 25 mL/min)	1500 mL in 2½ hours
24–26 kg	2500 mL	800 mL in 30 mins	(≈ 25 mL/min)	1700 mL in 2½ hours
27–29 kg	2800 mL	800 mL in 30 mins	(≈ 25 mL/min)	2000 mL in 2½ hours
30–31 kg	3000 mL	1000 mL in 30 mins	(≈ 35 mL/min)	2000 mL in 2½ hours
32–34 kg	3300 mL	1000 mL in 30 mins	(≈ 35 mL/min)	2300 mL in 2½ hours
35–38 kg	3500 mL	1000 mL in 30 mins	(≈ 35 mL/min)	2500 mL in 2½ hours
39–43 kg	4000 mL	1200 mL in 30 mins	(≈ 40 mL/min)	2800 mL in 2½ hours
44–49 kg	4500 mL	1500 mL in 30 mins	(≈ 50 mL/min)	3000 mL in 2½ hours
50–54 kg	5000 mL	1500 mL in 30 mins	(≈ 50 mL/min)	3500 mL in 2½ hours
≥ 55 kg	6000 mL	2000 mL in 30 mins	(≈ 70 mL/min)	4000 mL in 2½ hours

Infants (under 1 year, plan C 100mL/kg in **6 hours**)

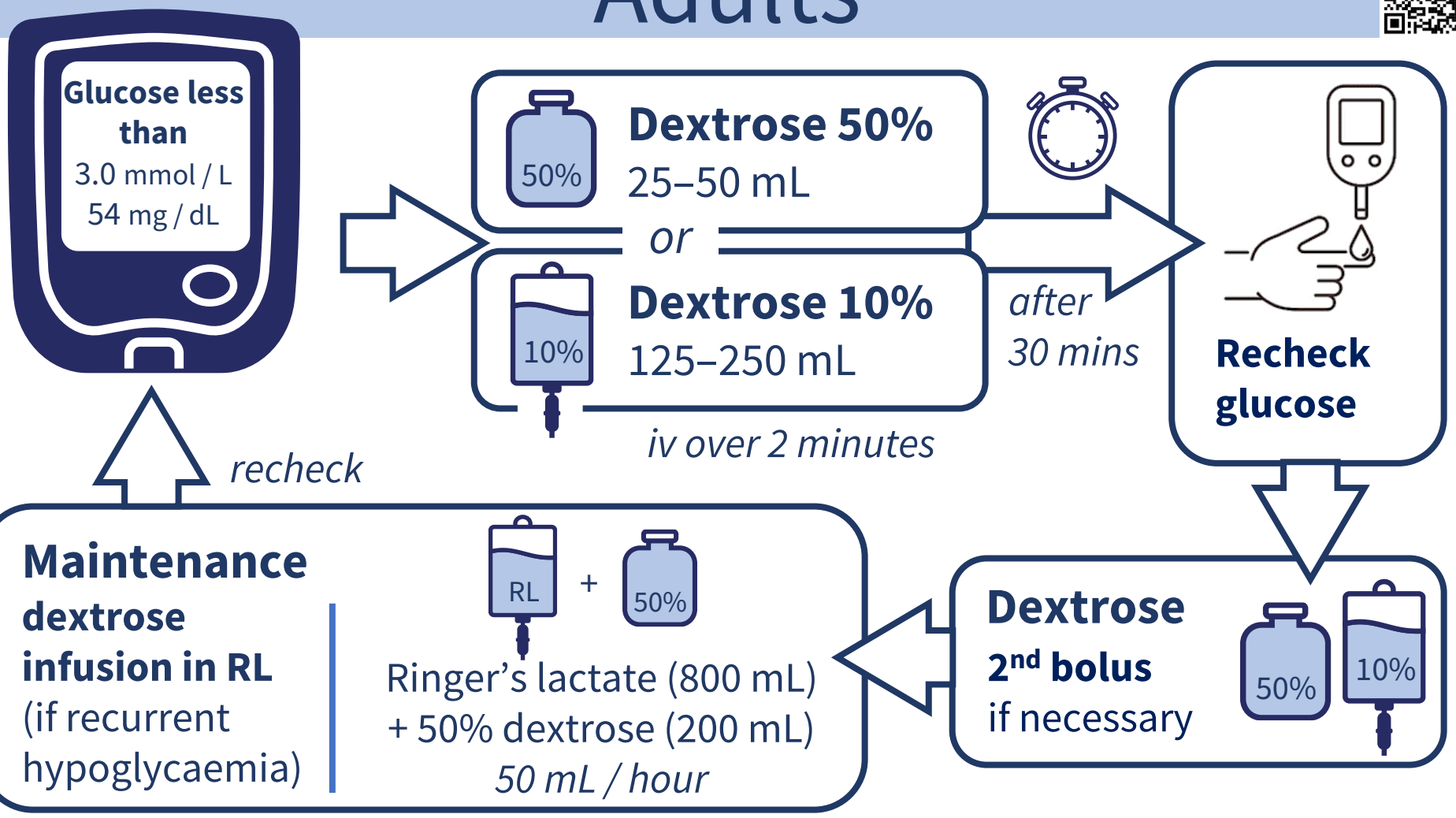
Age (years)	Weight (kg)	Total volume	30 mL/kg* Ringers over 1 hour	70 mL/kg* Ringers over 5 hours
<1 month	3–4 kg	350 mL	105 mL in 1 hour	245 mL in 5 hours
1–2 months	4–5 kg	450 mL	135 mL in 1 hour	315 mL in 5 hours
2–3 months	5–6 kg	550 mL	165 mL in 1 hour	385 mL in 5 hours
3–4 months	6–7 kg	650 mL	195 mL in 1 hour	455 mL in 5 hours
4–7 months	7–8 kg	750 mL	225 mL in 1 hour	525 mL in 5 hours
7–10 months	8–9 kg	850 mL	255 mL in 1 hour	595 mL in 5 hours
10–12 months	9–10 kg	950 mL	285 mL in 1 hour	665 mL in 5 hours

**Fluid should be administered through an appropriate infusion set (1 mL = 60 drops; 150 mL chamber for paediatrics)*

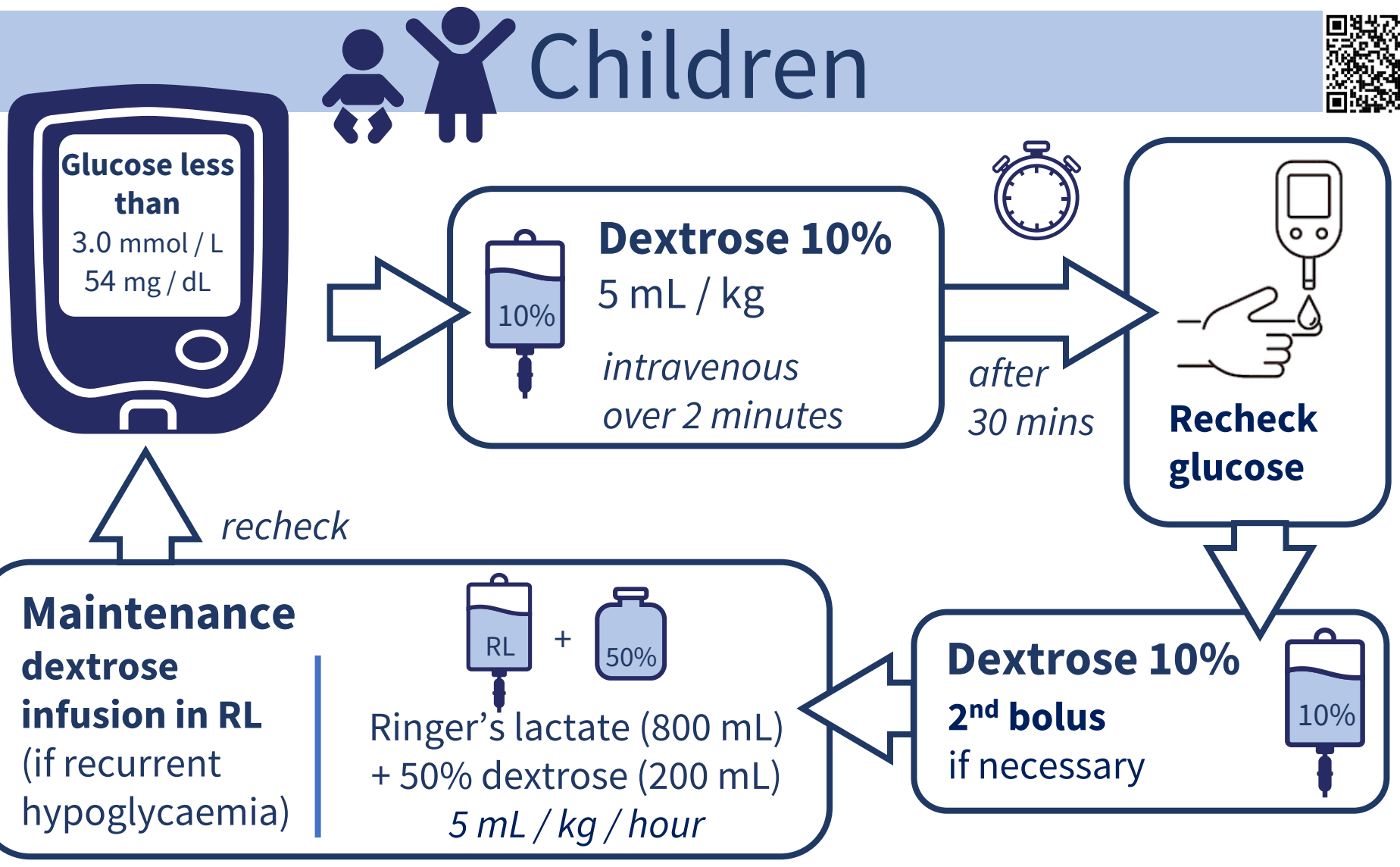
World Health Organization

Hypoglycaemia quick reference

Adults



Children



If no intravenous access, 2-5ml 50% dextrose fluid can be placed directly the cheek using a syringe. Do not use a needle or spoon.

References (scan QR code)

1) *WHO-ICRC Basic Emergency Care*
2) *Pocket book of hospital care for children*

World Health Organization

Clinical tools for cholera treatment facilities
v1.0

Discharge assessment

- ❑ Ensure patients and their caregivers understand that anyone with at least 3 episodes of diarrhoea in 24 hours should seek treatment.
- ❑ Ensure all patients and their caregivers understand how to prevent cholera before discharge.
- ❑ Provide at least 4 packets of ORS for all patients who are discharged to use at home.

  The packets can also be used for any family members or neighbours how begin to have diarrhoea.
- ❑ For paediatric patients give sufficient zinc to complete 10 days treatment.

Case definition

Chikungunya Fever should be suspected when a person develops sudden onset of fever, joint manifestations and rash.

Cases are to be categorized as:

a) **Possible case:** a patient meeting only clinical criteria
b) **Probable case:** a patient meeting both the clinical and epidemiological criteria
c) **Confirmed case:** a patient meeting the laboratory criteria, irrespective of the clinical presentation

Criteria for the Identification of Chikungunya Infection

Clinical criteria

- acute onset of fever >38.5°C
- severe arthralgia/arthritis not explained by other medical conditions

Epidemiological criteria

- Residing or having visited epidemic areas
- having reported transmission within 15 days prior to the onset of symptoms

Laboratory Criteria: at least one of the following tests in the acute phase:

- Virus isolation by Cell Culture
- Presence of viral RNA by real Time RT-PCR (Within 5 days of onset of Illness)
- Presence of viral specific IgM antibody in single serum sample collected within 5 to 28 days of onset Fever
- Four-fold Rise of IgG antibody in samples collected at least three weeks apart (1st sample after 7 days)

Clinical Presentation

Table 1: Clinical features of Chikungunya fever

Common	**Infrequent**	**Rare in adults but seen in children**
Fever Arthritis/ Arthralgia Backache Headache Rash	Stomatitis Oral ulcers Exfoliative dermatitis Photosensitive hyperpigmentation	Photophobia Retro-orbital pain Vomiting Diarrhoea Mental confusion Signs of meningeal irritation

Fever

The fever varies from low grade to high grade, lasting usually for 24 to 48 hours. Fever rises abruptly in some, reaching 39-40^{0}C, with chills and rigor, no diurnal variation, usually subsides with use of antipyretics.

Joint manifestation

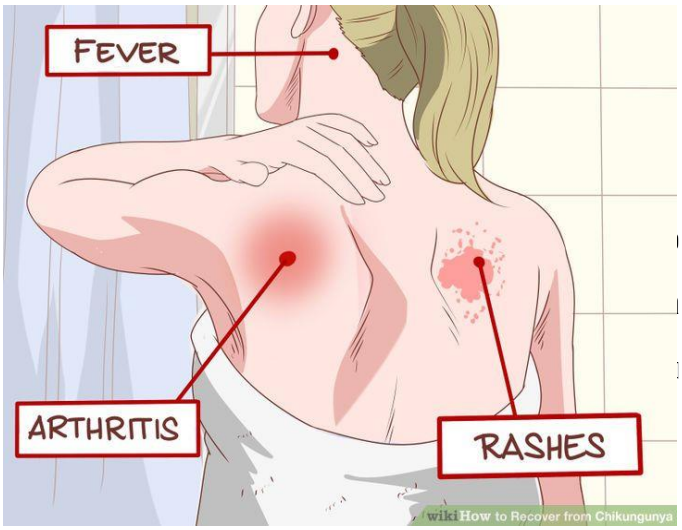
The joint symptoms usually start with arthralgia or arthritis. Involvement is symmetric and often ankles wrists and small joints of the hand are the worst affected. Migratory polyarthritis with effusions may be seen, but resolves in the majority. Larger joints like knee and shoulder and spine were also involved. Pain tends to be worse in the morning, relieved by mild exercise and exacerbated by aggressive movements. The pain may be relieved for 2-3 days and then reappear in a saddle back pattern. There is a tendency for early and more significant involvement of joints with some trauma or degeneration.The classical bending phenomenon was probably due to the lower limb and back involvement which forced the patient to stoop down and bend forward.

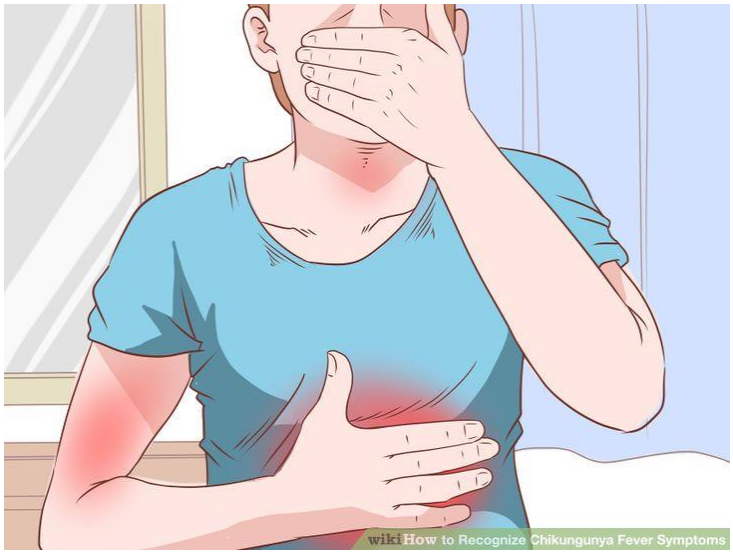

wikiHow to Recognize Chikungunya Fever Symptoms


103.5
wikiHow to Recognize Chikungunya Fever Symptoms


wikiHow to Recognize Chikungunya Fever Symptoms


wikiHow to Recover from Chikungunya


FEVER
ARTHRITIS
RASHES
wikiHow to Recover from Chikungunya


wikiHow to Recognize Chikungunya Fever Symptoms

in children. Most skin lesions recovered completely except in cases where the photosensitive hyperpigmentation persisted. Stomatitis and oral ulcers also observed.

Neurological manifestation

Neurologic manifestations of meningo-encephalitis, encephalitis, acute encephalopathy Guillain-Barré syndrome and myelitis were reported.

Ocular manifestation

Neuro-retinitis and uveitis in one or both eyes may occur. The main ocular manifestation associated with the recent epidemic outbreak of chikungunya virus infection in South India included granulomatous and nongranulomatous anterior uveitis, optic neuritis, retrobulbar neuritis, and dendritic lesions. Majority of the patients recover with good vision.

Clinical Classification of Chikungunya Fever

Chikungunya fever is classified in to three categoriesbased on severity of clinical presentation.

Mild

- Low grade fever
- Mild artharlgia
- Mild focal myalgia
- General weakness
- Skin rash/itching

Moderate

- Low to high grade persistent fever
- Moderate arthralgia /arthritis
- Generalized myalgia

- Hypotension
- Mild bleeding
- Retro-orbital pain
- Oliguria

Severe

- Persistent high grade fever
- Severe arthralgia/Arthritis
- Persistent vomiting / Diarrhoea
- Altered sensorium
- Bleeding (GI bleeding due to use of drugs e.g. analgesics)
- Shock due to persistent vomiting and diarrhoea

High Risk group

Chikungunya infection with one of the following conditions may be considered as high risk.

a) Co-morbid condition

Hypertension, Diabetic, CAD/CVD, Geriatric age, Pregnancy, COPD,

b) Co-infection

Dengue, Tuberculosis, Enteric fever, Pneumonia, HIV, Malaria

Clinical course and outcome

- Acute symptoms typically resolve within 7–10 days
- Some patients might have relapse of rheumatologic symptoms (e.g., polyarthralgia, polyarthritis, tenosynovitis) in months following acute illness

- In variable proportions of patients, joint painmay persist for months to years.
- Rare complications include
    - uveitis, retinitis, myocarditis, hepatitis, nephritis, bullous skin lesions, hemorrhage, meningoencephalitis, myelitis, Guillain-Barré syndrome, and cranial nerve palsies may develop

Differential diagnosis

Some of the diseases which can be considered in differential diagnosis are:

(1) **Dengue fever**: Severe backpain with purpuras or active bleeding might suggest dengue fever. Confirmatory laboratory diagnosis is possible.

(2) **Reactive Arthritis**: In general, any arthritis that follows a febrile gastrointestinal or genitourinary infection (triggering microbes) is considered a reactive acute inflammatory arthritis if it lasts less than six months. The hallmark feature is enthesitis where collagenous structures such as tendons and ligaments insert into bone are involved. Oral mucosal ulcers are seen.

(3) **Serum sickness illness**: Polyarthritis may be associated with a serum sickness type reaction caused by vaccine, medication or other viral infections

(4) **Rickettsial disease** can present with fever, rash and joint pains. Confirm by serology.

(5) **Rheumatic fever**: More common in the children and presents with fleeting (migratory) polyarthritis predominantly affecting the large joints. Modified Jones criteria should be the basis for diagnosis. Raised ASO titre and a history of recurrent sore throat are other points to be noted.

(6) **Malaria**: patient can present with high fevers and may also complain of joint pains. Periodicity of fever and alteration of consciousness / seizures should prompt a diagnosis for malaria

(7) **Leptospirosis**: Severe myalgia localized to calf muscles with conjunctival congestion/ or subconjunctival haemorrhage with or without oliguria or jaundice in a person with history of skin contact to contaminated water would suggest Leptospirosis

Clinical and laboratory features of chikungunya virus infections compared with dengue virus infections

Chikungunya		**Dengue**
Fever (>38.5°C)	+++	++
Arthralgia	+++	+/-
Arthritis	+	-
Headache	++	++
Rash	++	+
Myalgia	+	++
Hemorrhage	+/-	++
Shock	-	+
Lymphopenia	+++	++
Neutropenia	+	+
Thrombocytopenia	+	+++
Hemoconcentration	-	++

Chikungunya and dengue

- Difficult to distinguish chikungunya and dengue based on clinical findings alone
- Chikungunya and dengue viruses transmitted by the same mosquitoes

- The viruses can circulate in the same area and cause occasional co-infections in the same patient
- Chikungunya virus more likely to cause high fever, severe polyarthralgia, arthritis, rash, and lymphopenia
- Dengue virus more likely to cause neutropenia, thrombocytopenia, hemorrhage, shock, and death
- **Patients with suspected chikungunya should be managed as dengue until dengue has been ruled out**
- Proper clinical management of dengue reduces the risk of medical complications and death
- Aspirin and other NSAIDs can increase the risk of hemorrhage in patients with dengue

Principles of Management

Mild and moderate cases can be managed at home. Severe cases should be managed at hospital.

A. Home Management

- Consume plenty of water with electrolytes (approximately 2 litres of home available fluids with salt in 24 hours).
- Take paracetamol tablets during periods of fever (up to 1000 mg tablets four times daily), in persons with no preexisting liver or kidney disease. Children may be given 50-60 mg per kg body weight per day in divided doses.
- Adequate rest in a warm environment
- Cold compresses may help in reducing joint damage. Heat may increase/worsen joint pain and is therefore best to avoid during acute stage.
- Refrain from exertion. Mild forms of exercise and physiotherapy are recommended in recovering persons.

- Avoid self medication with aspirin or NSAIDs.
- Antihistamines can be used for itching.

B. Admission criteria

1. If the person is hemo-dynamically unstable (frequent syncopal attacks, hypotension with a systolic BP less than 90 mmHg or a pulse pressure less than 30 mmHg),
2. oliguria (urine output less than 500 ml in 24 hours),
3. altered sensorium
4. bleeding manifestations
5. persons not responding or having persistent joint pain or disabling arthritis even after three days of symptomatic treatment
6. persons above sixty years
7. infants (below one year of age)
8. Pregnancy
9. High risk Group

C. Hospital management

- Assess for dehydration and institute proper rehydration therapy. (5% DNS, Normal Saline, & baby saline for paediatric cases)
- Take paracetamol tablets during periods of fever (up to 1000 mg tablets four times daily), in persons with no preexisting liver or kidney disease. Children may be given 50-60 mg per kg body weight per day in divided doses.
- Antibiotics can be used to treat secondary bacterial infection.
- Antihistamines can be used for itching.
- Cold compresses may help in reducing joint damage.Heat may increase/worsen joint pain and is therefore best to avoid during acute stage.
- Refrain from exertion.

- Mild forms of exercise and physiotherapy are recommended in recovering persons.
- Aspirin or NSAIDs should be avoided during first 10 days. NSAIDs can be used in Chikungunya only when Dengue fever is adequately excluded.
- **C**utaneous manifestations with topical or systemic drugs, and neuro-psychiatric problems with specialist care and drugs).
- Steroid has no role in acute stage.
- In cases with ophthalmic symptoms complications, consult with ophthalmologists.

Treatment of serious complications

- Bleeding disorders with blood components- platelet transfusions in case of bleeding with platelet counts of less than 50,000 cells per cu mm., fresh frozen plasma, or Vitamin K injections if prothrombin time INR is more than 2.
- Hypotension with fluids/ inotropics, acute renal failure with dialysis, contractures and deformities with physiotherapy/surgery,
- Patients with myo-pericarditis or meningoencephalitis may require intensive care with regular monitoring, inotropic support, ventilation etc.
- Anti-neuralgic drugs (Amitryptyline, Carbamazepine, Gabapentin, and Pregabalin) can be used in standard doses in disturbing neuropathies.
- Use hydroxychloroquine 200 mg orally once daily or chloroquin phosphate 300 mg orally per day for a period of four weeks in cases where arthralgia is refractory to other drugs.
- Hyperpigmentation and papular eruptions can be managed with Zinc oxide cream and/or Calamine lotion

Special Situation

Effect on pregnancy

Chikungunya fever **appears to have a direct impact on pregnancy** with rare reports of spontaneous abortions and mother-to-child transmission in perinatal period.

Effect on neonates

Mothers afflicted with Chikungunya fever in the perinatal period (−4 days to +1 days before/after delivery) can transmit Chikungunya fever to neonates by vertical transmission. Caesarean section does not appear to prevent transmission. Neonatal CHIK fever is associated with fever, poor feeding, pain, distal edema, various skin manifestations, seizures, meningo-encephalitis, and echocardiographic abnormalities in the newborn. No infants are found to be infected with chikungunya virus through breast feeding. Mothers are encouraged to breast feed their infants.

Laboratory Diagnosis

Preferred Specimen:

1) Blood or Serum
2) CSF (if neurological features present)

Routine Laboratory Investigations

- CBC
    - Leucopenia (Decreased WBC)
    - Thrombocytopenia: rare
    - ESR: Usually Elevated
- C-Reactive Protein: Increased during the Acute Phase and may remain elevated for a week

- SGPT: Elevated

Specific Laboratory Investigations: at least one of the following tests in the acute phase:

- Virus isolation by Cell Culture
- Presence of viral RNA by real Time RT-PCR (Within 5 days of onset of Illness)
- Presence of viral specific IgM antibody in single serum sample collected within 5 to 28 days of onset Fever
- Four-fold Rise of Ig G antibody in samples collected at least three weeks apart (1st sample after 7 days)

N.B.:

1. Diagnostic validity of currently available ICT kit for the detection of IgM and IgG for Chikungunya virus is very poor.
2. RT-PCR for Chikungunya is performed only at IEDCR.

Public Health Measures

Patient when infected can spread the infection by spreading the infection through mosquitoes. It is important to break this transmission by minimizing the vector density by community participation and taking appropriate control measures in the hospital setting by following measures.

- Minimizing transmission of infection: This can be done in the following ways:
    - Risk communication to the household members
    - Minimize the vector population
    - Minimize the vector-patient contact (Aedes mosquitoes bite during day time, mostly in the morning and late afternoon)